Detecting Invisible Keypoints

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***Abstract*— The ability to visually localize object keypoints can provide robots with a functional understanding of their surrounding world. However, keypoints may often be hidden from the egocentric perspective of the robot, e.g., due to self-occlusion, occlusion by other objects, or falling outside the camera view. However, due to the inherent difficulty of the annotation task, existing benchmarks only provide a coarse keypoint representation and omit all out-of-view keypoints. Towards providing robots with a more useful representation of their surroundings under interactive and safety-critical scenarios, this work introduces the first large-scale benchmark for fine-grained invisible keypoint recognition and localization. We focus on the specific case of dense urban settings with vehicles and leverage the benchmark to analyze modeling challenges within the novel task. Our simulation settings build upon the open-source CARLA environment to incorporate 62 vehicle keypoints (e.g., handles, license plate, wheels, rear lights). We then employ a state-of-the-art keypoint detection model and explore its performance across various experimental factors, including varying camera perspectives, significant object occlusion, and harsh ambient conditions. Given the novel analysis, we further introduce a model training framework for learning more robust and geometry-aware keypoint models. Our findings and the introduced simulation environment facilitate functional and fine-grained robotic perception for further downstream tasks, from geometrical 3D reasoning to prediction, collision avoidance, and decision-making.**

Fig. 1: **Fine-Grained Detection of Occluded and Truncated Vehicle Keypoints in Images.** We introduce vehicle CAD models with annotated keypoints into CARLA to produce the first large-scale benchmark for studying the task of fine-grained 2D object keypoints detection under dense dynamic agents with common occlusion and truncation. The benchmark is used to develop a framework for training perception models with improved geometrical reasoning. An example frame from the introduced dataset is shown with vehicle keypoints that are out of the camera view. Connecting segments are color-coded based on their corresponding side of the vehicle.

I. Introduction

The ability to reason over unseen components of scenes is often essential to safely interact with the surrounding world. Humans leverage a geometric understanding of the world in order to reason over what may not be fully visible. For instance, humans are able to effortlessly infer the 3D position of a door handle of a vehicle, even if it is not directly visible from their perspective. In dense urban areas, human drivers can also estimate the complete 3D structure of surrounding objects from small visible portions, e.g., a car's bumper or window. Object truncation often occurs at close proximity to the observer, as occluded vehicles could suddenly emerge at close proximity to the ego-vehicle affording little time to react [1, 18, 25]. However, the performance of current perception models degrades under occlusion [19, 24, 26, 26, 26, 39, 40] and is rarely studied under severe truncation due to the lack of suitable benchmarks [17]. Particularly for truncation scenarios, researchers and developers often attempt to circumvent the challenging geometrical perception task with additional sensors, e.g., multiple cameras or LiDARs with an expanded field of view. While effective, this can limit the scalability of algorithms to use-cases where this may not be possible, such as more minimal and cheap robotic or wearable systems [12, 13, 21, 23]. For instance, a smartphone-based [23] or a low-profile ground system [12, 13, 21] may frequently observe objects that partially fall outside of its field of view.

Despite the increase in robot perception benchmarks in recent years, a complete keypoint representation of objects is often lacking in existing benchmarks [7, 15, 20, 29, 38], with the non-visible keypoints only partially annotated or missing all together [26, 31]. This is due to the difficulty in accurately annotating occluded or truncated keypoints. Moreover, partly visible objects are rare in most existing datasets [26]. Several recent works have tackled occluded keypoint detection [2, 31], however, only annotations for fully visible keypoints have been made publicly available. In addition, prior work may annotate only a coarse keypoint representation [26] which lacks many semantic components and has limited utility for further downstream tasks, e.g., mesh reconstruction [17] or decision-making. While synthetic benchmarks for robotics have gained popularity in recent years [6, 28, 32], current environments, such as CARLA [6], do not support keypoint annotations for vehicles. In this work, we address this current gap in benchmarks by introducing keypoint-

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annotated 3D vehicle models. The vehicles can then be spawned densely in the simulation, together with diverse cameras and environmental conditions, to generate a large-scale dataset used in our analysis. Performing data collection in simulation also has the added benefit of being highly controllable, we were able to generate data across different urban towns and weather conditions varying in lighting and visibility. Towards robust and functional perception models, we leverage the novel dataset to re-visit current challenges and properly benchmark model development. To demonstrate the challenging nature of our task, we perform comprehensive experimentation with a state-of-the-art model based on vision transformers [41]. By analyzing its performance over visible, obstructed, and truncated keypoints, we are able to analyze challenges and uncover research opportunities, e.g., through appropriate loss and training process design.

Our Contributions: Towards exploring the problem of geometrical object understanding with partial visibility, our contribution is threefold: (1) We introduce the Invisible Keypoint (INK) detection benchmark with fine-grained annotation including frequent partially visible instances. We collect realistic videos of dense urban road scenarios in simulation, incorporating images across varying perspectives, towns, and weather conditions (e.g., rain, sun glare). Our introduced dataset is an order of magnitude larger than existing vehicle keypoint datasets [15, 26, 31, 36, 38] with over 100k fully annotated frames and over 440k vehicle instances. (2) The comprehensive benchmark enables, for the first time, a rigorous analysis of challenges with current state-of-the-art keypoint detection methods for our task. To gain further insights and guide future researchers, we are also able perform targeted analysis over varying keypoint visibility and type. (3) To further understand and address the difficult learning task, we analyze the role of an additional geometric loss, trained via a gradual training process, which we demonstrate to consistently improve model performance under conditions with severe occlusion and truncation. The manually annotated 3D vehicle CAD models, collected benchmark, and keypoint detection models will be made publicly available to facilitate future research on our less studied but challenging reasoning task.

II. Related Work

Our work builds on recent advances in virtual benchmarks for robotics. We also build on state-of-the-art keypoint detection models based on vision transformers, however, we note these are often evaluated on benchmarks with rare occlusion and truncation.

Keypoint Detection: Datasets and methods for keypoint detection have been long studied in the robotics community. Until recently, the task of keypoint detection has been mostly studied using R-CNN-based approaches [11, 14, 16, 27, 33]. However, with the advent of vision transformers [5], existing transformer-based methods were extended to accommodate 2D feature maps. Such patch-based attention approaches have shown promising performance improvements over purely convolutional architectures. Consequently, transformer-based approaches are being increasingly used for the keypoint recognition and localization task [5, 34, 41]. We, therefore, employ the state-of-the-art ViTPose [41] model in order to uncover current challenges and failure modes in model performance under severe occlusion and truncation cases. We note that both model types are not generally evaluated on truncated instances.

Detection under Occlusion and Truncation: While detection and tracking of occluded and truncated objects have long been studied (e.g., [19]), these still pose a significant challenge to perception systems today. Specifically for the lesser studied case of truncated instances, Vedaldi and Zisserman [35] developed a structured output model for accounting for partial truncation. The model can be used to infer boxes that are outside the camera view and more aligned with the physical dimensions of objects. Several subsequent works leverage alignment between objects (and their keypoints) with 3D models of vehicles [2, 15, 31]. Patterns of real-world occlusion scenarios may generally provide some context for a predictive model, e.g., a vehicle partially hidden behind another vehicle [37]. There have been several recent attempts to reduce the effect of these truncated and occluded keypoints. Reddy et al. [26] employs a graph neural network to explicitly predict occluded 2D and 3D keypoints. A lifting process of 3D structured keypoints from 2D keypoints has been shown to result in improvements in predicting occluded keypoints [3, 30, 44, 45]. However, truncation can introduce a significant additional challenge due to the lack of context and complete reliance on object geometry modeling. Moreover, many of the aforementioned studies leverage a coarse keypoint representation in less diverse settings compared to the ones introduced by our benchmark. Yet, fine-grained keypoint detection tasks can provide more functional perception for safe and smooth robot navigation and interaction. We also note that by formulating our task as a 2D image-based keypoint detection task, our models can be used to inform subsequent 3D keypoint prediction tasks (e.g., [2, 15, 31]).

Synthetic Datasets: Real-world datasets are often restrictive, i.e., in terms of size and diversity, particularly when involving a cumbersome annotation task. Moreover, most real-world datasets such as KITTI [2, 10] or ApolloCar3D [31] contain very few truncated instances. With modern realistic rendering engines, photorealistic simulations can provide ample data for training and validating robotic perception models. Pedestrian keypoints have been studied on several synthetic benchmarks (e.g., JTA-dataset [9], VIPER [28], MOTSynth [8]). While related to our task, vehicles are larger objects which can further benchmark the ability of models to reason over geometry, i.e., even with large areas of the object outside the view of the camera. We also provide a comprehensive keypoint representation, beyond most prior work (62 keypoints). This ensures learning functional scene understanding models for subsequent tasks, e.g., amodal mesh generation [17]. Most related to our work is the Apol-

TABLE I: **The Invisible Keypoints (INK) Benchmark.** We introduce a comprehensive benchmark for analyzing challenges in detecting invisible object keypoints in images. The large synthetic video dataset contains dynamic and dense vehicles across harsh generalization conditions, including camera perspectives, cities, and weathers. By navigating in a virtual world, we are able to collect accurate keypoints under severe occlusion and truncation (i.e., out of the camera view). We note that ApolloCar3D did not annotate occluded or truncated keypoints nor released fitted 3D car models to the visible keypoints.

Dataset	# Annotated images	Source	# Keypoints	# Cities	# Weathers	Avg. Cars/Image	Video	Occlusion	Truncation
PASCAL VOC [15]	500	Real	20	-	-	-	✗	✗	✗
ObjectNet3D [38]	7,345	Real	14	-	-	1.75	✗	✗	✗
Veri776 [36]	50,000	Real	20	1	1	1	✗	✗	✗
CarFusion [26]	53,000	Real	12	1	1	1.89	✓	✓	✗
ApolloCar3D [31]	5,277	Real	**66**	-	-	**11.7**	✓	✗	✗
INK (This Work)	**102,667**	Simulation	62	**5**	**14**	4.29	✓	✓	✓

loCar3D [31] benchmark, which also extensively annotates keypoints of cars in traffic scenes (up to 66 keypoints). However, only visible keypoints are annotated and provided with the dataset. Occluded and truncated keypoints have not been annotated. Moreover, while 3D car model fitting can be employed to recover missing keypoints [31], these have not been released with the public dataset. Moreover, the diversity of towns and weathers is constrained, and truncated instances are rare in the benchmark. Thus, our work complements several recent efforts in the analysis of vision models in simulation. Our proposed INK dataset also contributes in terms of generalization, i.e., evaluation of fine-grained keypoint representation over diverse image perspectives and cluttered environmental settings.

III. INK Benchmark

Motivated by the lack of large-scale benchmarks with fine-grained keypoint annotations (Table II), we sought to provide a standardized benchmark, collected during dense urban navigation scenarios with frequent object occlusion and truncation. This section describes our annotation and benchmark generation process.

Keypoint Annotation of Vehicle Models: We leverage the widely used open-source CARLA simulation [6] to collect a diverse dataset across perspectives, weather, and cities. However, CARLA does not currently provide keypoint information for vehicles. Thus, to create our dataset we manually annotate the 3D CAD models. Annotating each model can take up to several hours. By releasing the annotated models in addition to the dataset, we hope to facilitate future studies in keypoint-based perception and learning, e.g., for imitation and reinforcement learning [22, 43]. We manually annotate 62 keypoints for each vehicle model in Unreal Engine 4. The models included vehicles with two doors and four doors. We then spawn the vehicles in the simulation to generate large amounts of data.

Data Collection Process: In order to support diverse scenarios and robotic use cases, we leverage eight camera perspectives during data collection in the simulation. While we focus on image-based detection analysis, we also collect additional sensor data, e.g., instance segmentation and depth, for future research to be released as part of the benchmark. We collect data from both the perspective of an on-road ego-vehicle and sidewalk walkers (from each, we collect four camera views—front, back, left, and right). We label each keypoint for visibility, occlusion, and truncation, i.e., out of the bounds of the camera. In the case of the latter, very few benchmarks exist today with accurate ground truth, i.e., due to the challenge in the manual annotation of such keypoints. However, many safety-critical situations could occur at close proximity when an object is partially out of the camera view. Even with wide-angle cameras, occlusion and truncation can occur frequently in the real world. For instance, the camera lens itself could become covered due to dirt, obstruction, or malfunction. We further note that our data contains full 3D information for each keypoint. While our dataset and model in Sec. IV can be potentially generalized to output 3D predictions, we leave such a task for future work. Moreover, our 2D detection task can be used to guide 3D prediction (e.g., as in Chen et al. [3]) in the future. Other downstream tasks in perception and interaction, e.g., sensorimotor navigation [6], are also beyond our scope and could be studied in the future.

Dataset Split: With the goal of model generalization evaluation, we partition the data into training and testing over towns and weather. We train over towns 1, 3, and 10 and using eight of the preset ambient (weather and time of day) conditions. We test over towns 2 and 5, as well as the remaining six preset conditions. This procedure results in 79,501 images for training and 23,166 for evaluation.

IV. Model

We sought to leverage our comprehensive dataset to understand challenges and opportunities in 2D-based keypoint detection models. We leverage the recent vision transformer-based model of Xu et al. [41] as our baseline (state-of-the-art on MS COCO). While we keep our model architecture simple, we explore the role of added geometry-based regularization as a training objective. We hypothesize that such objective can benefit reasoning over occlusion and truncation, as affirmed in Sec. V.

Vision Transformer: Following Xu et al. [41], our model incorporates a vision transformer [5] as the backbone in order to extract image features. A deconvolution layer-based

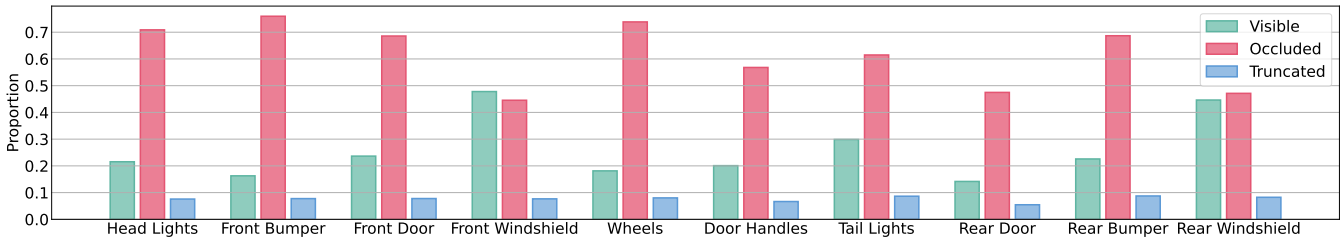

Fig. 2: **Dataset Visibility Statistics per Keypoint.** We cluster the 62 keypoints by types and show the distribution of visibility categories in our dataset.

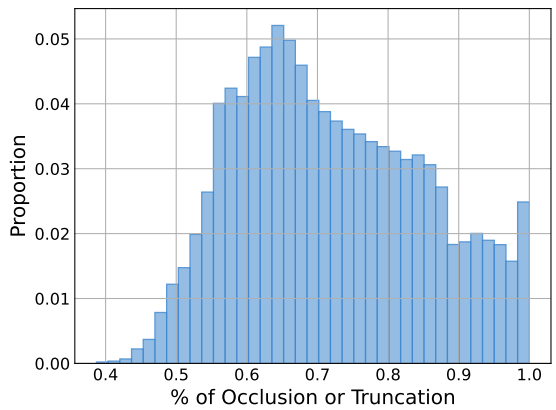

Fig. 3: **Vehicle Occlusion and Truncation Statistics in the Dataset.** We compute the percentage of overall keypoint occlusion or truncation per vehicle.

decoder is then used to estimate keypoint heatmaps. Given an input image $\mathbf{x} \in \mathcal{R}^{W \times H \times 3}$, the image is embedded into patch-based tokens [41]. Each token is represented as an array $\mathbf{z} \in \mathcal{R}^{\frac{H}{d} \times \frac{W}{d} \times C}$. Here $d = 16$ is a down-scaling ratio. We leverage 12 blocks of vision transformer as backbone. Each block is a basic structure of transformer encoder. An input feature is processed with (1) a Layer-Norm (LN), (2) a Multi-head self-attention layer (MSA) with 12 heads, and (3) a multi-layer perceptron (MLP) with two fully-connected layers.

Let $\mathbf{z_i}$ be the input of the $i-th$ transformer block, and let $\mathbf{z_{i+1}}$ be the output (and input to the next $(i+1)-th$ transformer block). We can then formally represent the computation of $i-th$ the transformer block as below,

$$\mathbf{z'_{i+1}} = \mathbf{z_i} \quad + \mathrm{MSA}(\mathrm{LN}(\mathbf{z_i})), \tag{1}$$

$$\mathbf{z_{i+1}} = \mathbf{z'_{i+1}} + \mathrm{MLP}(\mathrm{LN}(\mathbf{z'_{i+1}})), \tag{2}$$

Here, the multi-head self attention layer is computed over any input sequence of tokens $\mathbf{z}$

$$\mathrm{MSA}(\mathbf{z}) = \mathrm{softmax}\left(\frac{QK^T}{d_k}\right)V, \tag{3}$$

where $Q = \mathbf{z}W^Q$, $K = \mathbf{z}W^K$, $V = \mathbf{z}W^V$ and $W^Q \in \mathcal{R}^{d \times d_k}$, $W^K \in \mathcal{R}^{d \times d_k}$ and $W^V \in \mathcal{R}^{d \times d_v}$ are learned weight matrices.

We leverage an efficient decoder on top of the feature outputted by the backbone. To increase the output resolution, we directly up-sample the feature maps (by 4) using bilinear interpolation. Next, the features are processed with a convolution layer with a 3×3 kernel filter. The output $M \in \mathcal{R}^{\frac{H}{4} \times \frac{W}{4} \times N_m}$ is the heatmaps for keypoints as follows.,

$$M = Conv_{3\times3}(Bilinear(\mathbf{z})), \tag{4}$$

where N_m is the number of defined keypoints (62 in our case). We will publicly release all annotated vehicle models, collected benchmark, and trained detection models.

A. Loss Function and Model Training

We propose two key mechanisms for dealing with occlusion in truncation during model training. In addition to the common Mean Squared Error (MSE) regression loss computed directly over the predicted heatmaps M, we further introduce an auxiliary loss term, as described next.

Geometric Loss: We compute a geometric loss to regularize model training while also encourage inference of *geometrically plausible keypoint configurations*. Our key insight is to leverage a *soft-argmax function* [4] (over the heatmaps) in order to compute L2 distances directly between corresponding predicted keypoints location and ground truth (gt) location for each keypoint,

$$\mathcal{L}_{\text{geo}}(M) = \frac{1}{N_M} \sum_{j \in N_M} \|\mathrm{argmax}(M_j) - \mathrm{argmax}(M_j^{gt})\|_2^2 \tag{5}$$

where j indexes across keypoint types. The intuition for such a loss can be seen when comparing to the more commonly employed Chamfer loss. Given a set of predicted keypoint coordinates S, and a set of ground truth keypoint coordinates S_{gt}, one can obtain the Chamfer loss via

$$\mathcal{L}_{\text{Chamfer}}(S) = \sum_{p \in S} \min_{q \in S_{gt}} ||p-q||_2^2 + \sum_{q \in S_{gt}} \min_{p \in S} ||p-q||_2^2 \tag{6}$$

Thus, $\mathcal{L}_{geo}$ can be seen as a special case of the Chamfer loss where distances are computed between corresponding keypoint predictions and ground-truths. As the Chamfer loss identifies a nearest comparative point from the set of points to compute distances (i.e., not requiring pair correspondence), it does not require corresponding pairs of prediction and ground-truth. Yet, in our case of predicted 2D keypoints, we can simply use the loss in Eqn. 5. Thus, our final loss function becomes

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda \mathcal{L}_{\text{geo}} \tag{7}$$

Where $lambda$ is a scalar hyperparameter. Due to the increased complexity of the training objective and the overall task, we find that optimizing our model via a gradual training process benefits model performance, as described next.

Gradual Training: The geometrical loss $\mathcal{L}_{geo}$ can regularize training and result in a more geometrically consistent prediction, particularly under the instances of occlusion and truncation studied in our work. However, it can provide poor supervision at the onset of training, where the nearest keypoint match may often be a poor match. Thus, we propose a gradual training process, where we gradually increase the contribution of the geometric term to the overall loss in Eqn. 7. For ease of implementation, we first optimize the keypoint model for the MSE objective alone in our experiments. This step is then followed by introduction of the geometric loss term to obtain the final model trained with the joint objective. Based on our experiments, the gradual training process is critical to successful model training. Our baseline model is trained for a total of 70 epochs using AdamW [42] with a learning rate $5e^{-4}$ and $\lambda = 0.1$.

V. Experimental Evaluation

In this section, we perform three main experiments. First, report detection model failure modes over keypoint visibility level. Next, we introduce our proposed training framework in order to analyze its benefits for varying keypoint visibility levels. Third, we discuss a breakdown of model performance over keypoint categories (we group the original 62 keypoints into 10 categories). Given similar statistics in visibility, occlusion, and truncation among the keypoints (see Fig. 2), the breakdown enables drawing conclusions regarding the impact of the underlying geometry and keypoint appearance on model performance. Finally, we qualitatively analyze success and failure cases to contextualize the findings using concrete samples from the data.

Role of Keypoint Visibility: We begin by training the state-of-the-art ViTPose [41] model (using the standard MSE loss) on the INK benchmark. The results are shown in Table II. To explore the role of visibility in 2D object keypoint detection, we report the performance of the models over three levels of keypoint visibility by setting ignore flags over subsets of the data. As expected, we can see how model accuracy degrades under occlusion (e.g., from 62.6 AP to 54.1 AP). Under truncation, model performance degrades further, to 8.4 AP. This finding demonstrates truncation to be a significant failure mode. While the model is able to benefit from the occluding and surrounding scene context, this is not the case in truncated cases where extensive geometrical reasoning is required. Moreover, as truncated instances are rare in prior datasets, this sharp drop in performance highlights the need for more targeted development and evaluation over occluded and truncated instances.

Loss and Training Framework: In addition to heatmap-based loss used in standard keypoint detection, we propose to incorporate additional geometry-related supervision directly over keypoint positions. We incorporate the additional loss through two mechanisms, as shown in Table II. First, we incorporate the loss when training the model from scratch, i.e., using identical settings as the baseline ViTPose model. We also provide results using gradual training, where the model initially optimizes the standard MSE loss only and only afterwards the geometrical loss term is added. We hypothesize that the geometric loss may be too punishing at the onset of training, i.e., producing excessively large loss values due to the fact that the model has not yet fully learned object geometry. As the additional term requires increased geometrical reasoning, we find it to be beneficial to add the additional constraint later in the training. Table II depicts the benefit of this process, leading to consistent improvements in model accuracy over the baseline. Without gradual training, we find adding an additional loss term to slightly degrade performance. While we find the proposed model and geometrical reasoning to improve prediction accuracy over both invisible and visible keypoint detection, on truncated instances, performance increase from 8.4 AP to 9.7 AP. As truncated instances are the most challenging, even small gains in model accuracy are meaningful. Yet, modeling object geometry under heavy truncation requires further study in the future.

TABLE II: **Detection Results among Keypoint Visibility Categories.** We show a breakdown of detection performance based on keypoint types. Overall, we find that adding the geometric loss ('Geo') can improve model performance when accompanied by gradual training ('Gradual'), i.e., as a form of curriculum learning). The proposed training framework consistently outperforms the ViTPose [41] baseline. We also demonstrate truncated keypoints to involve a significantly more challenging perception task compared to occluded keypoints due to the lack of context.

Visibility	Model	AP	AP_{50}	AP_{75}	AP_M	AP_L
Fully Visible	ViTPose	62.6	73.5	61.6	70.2	78.6
	ViTPose + Geo	61.0	71.3	59.7	67.8	76.9
	ViTPose + Geo + Gradual	**65.1**	**75.0**	**64.2**	**71.3**	**84.6**
Occluded	ViTPose	54.1	63.8	52.8	64.6	76.1
	ViTPose + Geo	52.5	61.5	52.0	62.0	74.5
	ViTPose + Geo + Gradual	**56.5**	**65.3**	**55.5**	**65.4**	**82.8**
Truncated	ViTPose	8.4	16.6	6.6	**8.0**	11.9
	ViTPose + Geo	8.5	16.2	7.1	6.9	12.4
	ViTPose + Geo + Gradual	**9.7**	**17.1**	**8.4**	6.6	**14.8**
All	ViTPose	49.5	59.3	48.5	62.2	63.3
	ViTPose + Geo	47.9	56.9	47.1	59.9	61.9
	ViTPose + Geo + Gradual	**51.6**	**61.5**	**51.1**	**63.1**	**69.3**

Model Performance over Semantic Keypoint Groups: To understand whether keypoint type impacts detection performance (e.g., due to appearance or shape factors), we group the 62 annotated fine-grained keypoints into 10 groups based on vehicle parts. We then report model performance over each group, shown in Table III. In general, we observe that the model (both the ViTPose baseline and the proposed approach) performs worse on keypoints that belong to vehicle parts with a smaller area, such as tail lights and door handles. Moreover, the

TABLE III: **Results Breakdown for Different Keypoint Types.** We cluster the 62 keypoints into 10 semantically similar categories to analyze the benefit of the proposed framework.

Model	Keypoint Type	AP	AP_{50}	AP_{75}	AP_M	AP_L
ViTPose	Head Lights	56.0	62.1	55.8	68.2	70.4
	Tail Lights	48.4	51.6	48.3	60.3	61.8
	Front Bumper	66.1	75.3	65.6	79.8	78.2
	Rear Bumper	63.0	71.6	66.4	79.6	76.3
	Front Windshield	56.7	63.5	56.4	68.5	71.8
	Rear Windshield	67.1	76.7	66.4	80.4	81.0
	Front Door	58.3	64.9	57.9	70.6	72.2
	Rear Door	61.8	70.0	61.0	73.4	73.8
	Wheels	61.6	69.7	60.6	74.5	74.7
	Door Handles	49.7	52.7	49.9	60.9	63.6
ViTPose + Geo + Gradual	Head Lights	57.6	63.1	57.4	68.5	75.2
	Tail Lights	50.6	53.8	50.9	61.2	68.5
	Front Bumper	66.6	75.2	65.7	79.6	80.1
	Rear Bumper	66.8	72.9	68.4	80.3	79.0
	Front Windshield	58.4	64.7	57.7	68.9	76.6
	Rear Windshield	68.0	76.7	67.2	81.0	82.6
	Front Door	60.3	66.2	60.0	71.4	76.8
	Rear Door	64.2	71.2	63.9	75.2	77.5
	Wheels	62.9	70.1	62.5	75.0	77.8
	Door Handles	52.8	56.1	53.1	63.1	69.4

underlying pose distribution of objects in our application can also play a role with respect to the appearance characteristics of certain object parts.

Qualitative Results: We visualize ground truth and model prediction results for several samples in the dataset in Fig. 4 and Fig. 5. The figures depict how cases of occlusion are often resolved correctly by the model. Moreover, we can see how the model is able to handle small truncation, where most of the vehicle is still in view of the camera. However, as shown in Fig. 5, increasing truncation leads to general model failure. Here, the model must carefully reason over object orientation (and potentially scene layout) in a different portion of the image in order to correctly regress the out of view keypoints. We find this task to be beyond the capabilities of current models.

VI. Conclusion

While keypoint-based representations have been used for various robotic applications, few studies have isolated the role of occlusion and truncation on keypoint prediction. To address this current gap, we introduce a standardized benchmark based on the CARLA simulator. Moreover, to better reflect the challenging geometrical modeling task, we also proposed an improved loss function with a gradual model training mechanism. The analysis over different keypoint categories (i.e., varying over visibility and vehicle parts) was used to uncover current challenges and future research directions. We note that we particularly focus on a 2D detection task as it is useful for an array of subsequent tasks, from lifting and 3D vehicle model alignment and up to abstracting relevant semantic information for assistive and autonomous technologies. Even within the more simplified 2D detection task, we identify several critical failure modes in current models. We hope that the introduced benchmark and framework can inspire future researchers to explore more robust keypoint-based models in the future, e.g., by incorporating temporal and multi-modal information.

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Fig. 4: Successful cases of detection under occlusion and light truncation.

Fig. 5: Failure cases under more severe truncation conditions.

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